一、(1)、$\frac{\partial z}{\partial x}=y+f(\frac{y}{x})-\frac{y}{x}f'(\frac{y}{x})$

$\frac{\partial z}{\partial y}=x+f'(\frac{y}{x})$

(2)、$\frac{\partial u}{\partial x}=e^{xy}y^2(xy+1)$, $\frac{\partial u}{\partial y}=e^{xy}xy(xy+2)$

设$v=(\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}})$, $\frac{\partial u}{\partial v}(1,1)=\frac{\partial u}{\partial x}(1,1)\cdot\frac{1}{\sqrt{2}}+\frac{\partial u}{\partial y}(1,1)\cdot\frac{1}{\sqrt{2}}=\frac{5}{2}\sqrt{2}e$

(3)、设$F_1(x,y,u,v)=xu-yv-1$, $F_2(x,y,u,v)=yu+xv$, 则

$$\begin{pmatrix}\frac{\partial u}{\partial x} & \frac{\partial u}{\partial y}\\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y}\end{pmatrix}=-\begin{pmatrix}\frac{\partial F_1}{\partial u} & \frac{\partial F_1}{\partial v}\\ \frac{\partial F_2}{\partial u} & \frac{\partial F_2}{\partial v}\end{pmatrix}^{-1}\begin{pmatrix}\frac{\partial F_1}{\partial x} & \frac{\partial F_1}{\partial y}\\ \frac{\partial F_2}{\partial x} & \frac{\partial F_2}{\partial y}\end{pmatrix}=-\begin{pmatrix}x & -y\\ y & x\end{pmatrix}^{-1}\begin{pmatrix}u & -v\\ v & u\end{pmatrix}=-\frac{1}{x^2+y^2}\begin{pmatrix}xu+yv & -xv+yu\\ xv-yu & xu+yv\end{pmatrix}$$

$$\Rightarrow \frac{\partial u}{\partial x}=-\frac{xu+yv}{x^2+y^2},\ \frac{\partial u}{\partial y}=\frac{xv-yu}{x^2+y^2},\ \frac{\partial v}{\partial x}=\frac{-xv+yu}{x^2+y^2},\ \frac{\partial v}{\partial y}=-\frac{xu+yv}{x^2+y^2}$$

(4)、$\int_L e^{\sqrt{x^2+y^2}}ds=\int_0^{\frac{\pi}{2}}ae^a dt=\frac{\pi}{2}ae^a$

二、(1)、取$s_i=(\frac{1}{i},0)\neq(0,0)$, $t_i=(\frac{1}{i},\frac{1}{i})\neq(0,0)$, 则$\lim_{i\to\infty}s_i=\lim_{i\to\infty}t_i=(0,0)$, 而$f(s_i)=0$, $f(t_i)=\frac{1}{5}$

$\Rightarrow \lim_{(x,y)\to(0,0)}f(x,y)$不存在

$\Rightarrow f(x,y)$于$(0,0)$处不连续

(2)、$\frac{\partial f}{\partial x}(0,0)=\lim_{t\to0}\frac{f(t,0)}{t}=0$

$\frac{\partial f}{\partial y}(0,0)=\lim_{t\to0}\frac{f(0,t)}{t}=0$

三、(1)、$E^\circ=\{(x,y)\mid \frac{x^2}{4}-1<y<2-x\}$

$\overline{E}=\{(x,y)\mid \frac{x^2}{4}-1\le y\le 2-x\}$

E是闭区域

(2)、$E^\circ=\{(x,y)\mid x^2+y^2\neq1\}$

$\overline{E}=\mathbb{R}^2$

E既不是区域也不是闭区域

(3)、$E^\circ=\{(x,y)\mid 0<x^2+y^2<1\}$

$\overline{E}=\{(x,y)\mid x^2+y^2\le1\}$

E是区域

(4)、$E^\circ=\phi$

$\overline{E}=\{(x,y)\mid xy=1\}$

E既不是区域也不是闭区域

四、由对称性知 $\iint_D xy\,dxdy=0$，$\iint_D x^2y^2\,dxdy=4\iint\limits_{\substack{x+y\le 1,\\ x\ge 0,y\ge 0}} x^2y^2\,dxdy$

$$\Rightarrow I=4\iint\limits_{\substack{x+y\le 1,\\ x\ge 0,y\ge 0}} x^2y^2\,dxdy=4\int_0^1 x^2dx\int_0^{1-x} y^2dy=\frac{4}{3}\int_0^1 x^2(1-x)^3dx=\frac{1}{45}$$

五、证明：设P的坐标为(a,b)，则$A=\sqrt{a^2+b^2}$为P到O的距离，$\sqrt{(x-a)^2+(y-b)^2}$为P到D中某一点的距离，由$A>r$知$P\notin D$，那么$A-r\le\sqrt{(x-a)^2+(y-b)^2}\le A+r$

$$\Rightarrow \frac{m}{A+r}\le\frac{f(x,y)}{\sqrt{(x-a)^2+(y-b)^2}}\le\frac{M}{A-r}$$

$$\Rightarrow \iint_D\frac{m}{A+r}dxdy\le\iint_D\frac{f(x,y)}{\sqrt{(x-a)^2+(y-b)^2}}dxdy\le\iint_D\frac{M}{A-r}dxdy$$

$$\Rightarrow \frac{m}{A+r}\pi r^2\le\iint_D\frac{f(x,y)}{\sqrt{(x-a)^2+(y-b)^2}}dxdy\le\frac{M}{A-r}\pi r^2$$

□

六、设上述区域为Ω，由Guass公式及对称性知

$$I=\iint_\Sigma x^2dydz+y^2dzdx+z^2dxdy=2\iiint_\Omega(x+y+z)dxdydz=2\iiint_\Omega z\,dxdydz=2\iint\limits_{x^2+y^2\le 1}dxdy\int_{\sqrt{x^2+y^2}}^1 z\,dz$$

$$=\iint\limits_{x^2+y^2\le 1}(1-(x^2+y^2))dxdy=\int_0^{2\pi}d\theta\int_0^1(1-r^2)r\,dr=\frac{\pi}{2}$$

七、作一个以原点为中心，以充分小的$\varepsilon>0$为半径的圆周C_ε，正向为顺时针方向，设C_ε与Γ之间的区域为D，由Green公式，

$$\int_\Gamma\frac{ydx-xdy}{x^2+y^2}=-\int_{C_\varepsilon}\frac{ydx-xdy}{x^2+y^2}=-\frac{1}{\varepsilon^2}\int_{C_\varepsilon}ydx-xdy=-\frac{2}{\varepsilon^2}\pi\varepsilon^2=-2\pi$$

八、设$f(x,y)=x^2+y^2$，$F(x,y)=f(x,y)+\lambda(ax^2+2bxy+cy^2-1)=(a\lambda+1)x^2+2b\lambda xy+(c\lambda+1)y^2-\lambda$

令$\frac{\partial F}{\partial x}=2[(a\lambda+1)x+b\lambda y]=0$，①

$\frac{\partial F}{\partial y}=2[b\lambda x+(c\lambda+1)y]=0$ ②

$$ax^2+2bxy+cy^2=1\Leftrightarrow(x\ \ y)\begin{pmatrix}a & b\\ b & c\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=1$$

由于$\begin{pmatrix}a & b\\ b & c\end{pmatrix}$是正定阵，故特征值均大于0，因此该平面曲线是椭圆，为有界闭集。由于连续函数在有界闭集上必然取得最大值和最小值并且$x=0,y=0$不满足$ax^2+2bxy+cy^2=1$，因此上述方程组有非零解，故$\begin{vmatrix}a\lambda+1 & b\lambda\\ b\lambda & c\lambda+1\end{vmatrix}=(a\lambda+1)(c\lambda+1)-b^2\lambda^2=(ac-b^2)\lambda^2+(a+c)\lambda+1=0$

$$\Rightarrow \lambda=\frac{-(a+c)\pm\sqrt{(a-c)^2+4b^2}}{2(ac-b^2)}$$

由①可得 $\frac{y}{x}=-\frac{a\lambda+1}{b\lambda}$

由②可得 $\frac{x}{y}=-\frac{c\lambda+1}{b\lambda}$

$$\Rightarrow x^2+y^2=\frac{x^2+y^2}{ax^2+2bxy+cy^2}=\frac{\frac{x}{y}+\frac{y}{x}}{a\frac{x}{y}+2b+c\frac{y}{x}}=\frac{-\frac{c\lambda+1}{b\lambda}-\frac{a\lambda+1}{b\lambda}}{-a\cdot\frac{c\lambda+1}{b\lambda}+2b-c\cdot\frac{a\lambda+1}{b\lambda}}=\frac{(a+c)\lambda+2}{2(ac-b^2)\lambda+(a+c)}$$

$$=\lambda\cdot\frac{(a+c)\lambda+2}{2(ac-b^2)\lambda^2+(a+c)\lambda}=\lambda\cdot\frac{(a+c)\lambda+2}{-2[(a+c)\lambda+1]+(a+c)\lambda}=\lambda\cdot\frac{(a+c)\lambda+2}{-(a+c)\lambda-2}=-\lambda$$

$\Rightarrow x^2+y^2$ 的最大值为 $\frac{a+c+\sqrt{(a-c)^2+4b^2}}{2(ac-b^2)}$，最小值为 $\frac{a+c-\sqrt{(a-c)^2+4b^2}}{2(ac-b^2)}$

$\Rightarrow$ 平面曲线 $ax^2+2bxy+cy^2=1$ 上的点到原点的最远距离为 $\sqrt{\frac{a+c+\sqrt{(a-c)^2+4b^2}}{2(ac-b^2)}}$，最近距离为 $\sqrt{\frac{a+c-\sqrt{(a-c)^2+4b^2}}{2(ac-b^2)}}$